Notice, question, predict

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use observations to make a prediction that can be tested.

SUCCESS CRITERIA - I can...

1. Separate stated evidence from unsupported claims.
2. Extend a position pattern and state an assumption.
3. Name an observation that could test the prediction.

Core: teacher selects from 1-9. **Stretch:** 10 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Read a scale

A ruler shows a toy car at 2 cm, then at 7 cm. Draw both positions and the change. How far apart are the marks?

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $\Delta x = x_f - x_i$; $x_{next} = x_{now} + \Delta x$.
For a repeating position pattern only. i = initial; f = final. Check the pattern against evidence.

Worked reasoning: A cart is at 2, 5 and 8 m at 0, 1 and 2 s. Equal jumps of 3 m suggest 11 m at 3 s, if the pattern continues.

Your turn: A motion sensor recorded the first three positions in the table. Add the predicted position at 3 s. Draw the four positions on a line and distinguish the prediction from the measurements.

Time (s)	Position (m)
0	2
1	5
2	8
3	Predict: ____

Marzano check - circle: ① ② ③

03 Extract the evidence

A fictional space report says: "Repeated images show an object approaching Earth. Its path and size remain uncertain." Identify two statements the report supports and one conclusion it does not support. Explain the difference between an observation and a forecast.

Marzano check - circle: ① ② ③

04 Fit a simple model

A motion sensor tracks a cart at one-second intervals. Use the table to write a position rule, x(t), and predict x at 5 s. Explain the assumption that lets you extend the pattern.

t (s)	x (m)
0	1
1	4
2	7
3	10

Marzano check - circle: ① ② ③

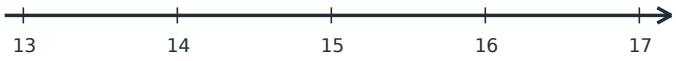
Notice, question, predict

antocciphysics

Continue down each column.

05 Challenge the prediction

At 5 s, the sensor measures 14.0 ± 0.2 m. Mark this interval and your prediction from question 4 on the number line. Do they agree? Give two possible reasons for any disagreement.



Position (m); right is positive

Marzano check - circle: ① ② ③

06 Design the observation

A telescope team can make two more observations of the fictional asteroid. Describe what to measure and when. Explain how the pair of measurements would improve a prediction of its future position.

Marzano check - circle: ① ② ③

07 Compare claims

Claim A: “A straight-line model predicts where it may be.” Claim B: “The asteroid will hit Earth.” Explain what additional evidence and physics separate these claims.

Marzano check - circle: ① ② ③

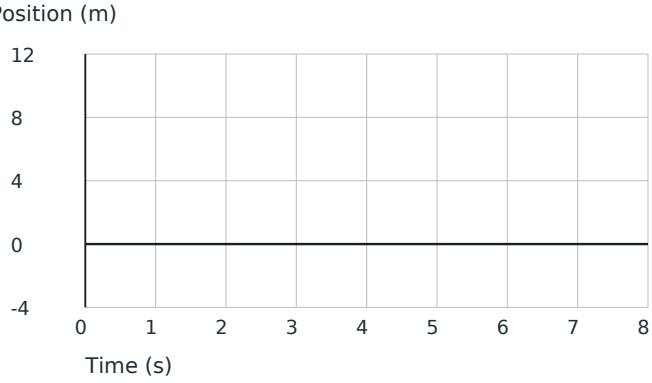
08 Choose useful evidence

Two groups time the same toy car over the same marked track. One records three times; the other writes "pretty fast." Which record would let someone check the claim? Explain.

Marzano check - circle: ① ② ③

09 Exit check

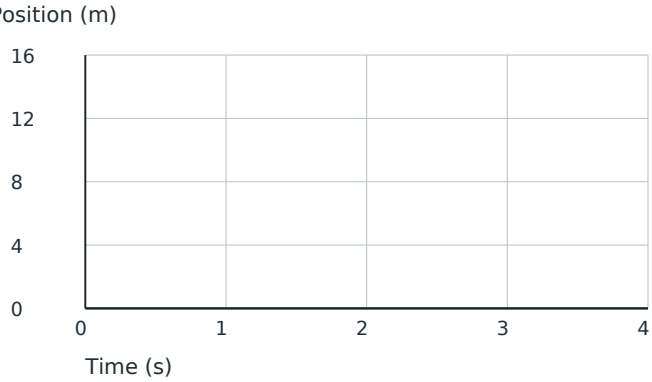
A robot's positions are 12, 8 and 4 m at 0, 2 and 4 s. Plot them, determine the velocity, and predict its position at 7 s. State an assumption and a later measurement that could test it.



Marzano check - circle: ① ② ③

10 STRETCH: test the model

Return to the worked example in problem 2. Construct two position rules that match all three observations in the worked case but disagree at 4 s. Explain why the data do not guarantee either extrapolation.



Marzano check - circle: ① ② ③